

WHY DOES SOMETHING EXIST RATHER THAN NOTHING?

An Information-Theoretic Answer via T_0 Theory

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ABSTRACT

The question "Why does something exist rather than nothing?" has remained unanswered for 2,500 years. We approach it through three independent lines of argument: (1) modern physics and quantum mechanics, (2) information theory, and (3) T_0 Theory (Theory of Volumetric Information in Time). We demonstrate that "nothing" is not an empty state but an information-theoretic potential. We show that physics forbids absolute emptiness, information theory shows that "nothing" cannot be described without contradiction, and T_0 Theory proves that potential must manifest. The conclusion from all three approaches is the same: "nothing" is a logical, physical, and informational impossibility. Science answers HOW. T_0 answers WHY. Together they show: there is only something.

"A seed cannot fail to sprout."

$1^{(\inf-1)/2n} = 1$ **Axiom.**

- **The Question and Why It Has No Answer Yet**

In 370 BC, Aristotle first asked: why is there something rather than nothing? In 1714, Leibniz formalized it as the fundamental question of metaphysics. In 1935, Heidegger called it "the fundamental question of philosophy." In 2026, it remains unanswered.

Why? Because every previous approach treated "nothing" as a valid starting state — an empty container waiting to be filled. Physics tried quantum foam. Philosophy tried necessary being. Theology tried creation ex nihilo. All failed for the same reason: they tried to explain how something emerged FROM nothing.

T_0 Theory takes a different approach: **nothing is not a valid information state.**

- **Three Independent Arguments Against "Nothing"**

- **Physics: The Vacuum Is Not Empty**

Modern quantum physics demonstrates that absolute emptiness is physically impossible:

Phenomenon	What it shows
Heisenberg Uncertainty	$\Delta E \cdot \Delta t \geq \hbar/2$ — energy is never exactly zero
Quantum vacuum fluctuations	Virtual particle pairs constantly appear and disappear
Casimir effect	Measurable force between plates in "empty" space
Zero-point energy	Even at absolute zero, energy remains

Conclusion from physics: A perfect vacuum is a theoretical limit, never achieved. Quantum laws forbid absolute zero energy. "Nothing" violates conservation laws.

- **Information Theory: "Nothing" Cannot Be Described**

Information theory reveals a deeper impossibility:

The Information Paradox: To record "there is nothing here" requires a medium. The medium is already something. Therefore "nothing" cannot be registered — and what cannot be registered does not exist in any information sense.

Shannon entropy $H = -\sum(p$

i

$\log p_i)$ equals zero only at maximum information (complete

certainty), not at "nothing." True emptiness would require $H = -\infty$, which is undefined. Kolmogorov complexity of "nothing" is not zero — describing it requires a language, which is already something.

Conclusion from information theory: "Nothing" is an information-theoretic impossibility. Any attempt to define it creates something.

- **T₀ Theory: NULL Is Potential, Not Emptiness**

T₀ Theory (Theory of Volumetric Information in Time) provides the deepest answer. In T₀, what we call "nothing" is not emptiness — it is NULL, the state of maximum potential:

NULL = not emptiness = potential = the state containing all possible states in
unmanifested form

This is the maximum entropy state in information theory: all possibilities equally probable, maximum potential information. Like a seed: it contains the entire tree, but has not yet expressed it.

• **The Mathematical Inevitability**

The simplest mathematical expression captures the entire argument:

$0^0 = 1$

In standard mathematics, 0^0 is called an "indeterminate form." In T

0

Theory, it is the
foundational equation of existence:

Symbol	Meaning
0	NULL = potential (unmanifested)
0	Self-reference: potential applied to itself
= 1	The first bit, the first manifestation

When potential applies itself to itself, it produces 1 — the first something. This is not a metaphor. It is the mathematical definition of what happens when you raise anything to the power of zero: you get the multiplicative identity, the unit. **NULL applied to NULL = the first bit.**

Zero density is unstable:

A state of zero information density (perfect nothing) is maximally unstable. Any information-theoretic system at maximum entropy is maximally sensitive to perturbation. No system can maintain perfect maximum entropy — the first fluctuation breaks the symmetry and produces a bit.

"This is not 'something came from nothing.' This is: a perfect nothing cannot be maintained."

• **The Formal Proof**

Step	Statement
Lemma 1	"Nothing" = NULL = maximum entropy potential (all possibilities open)
Lemma 2	Maximum entropy is maximally unstable — any fluctuation breaks symmetry
Lemma 3	The first fluctuation produces a bit: 0 or 1
Lemma 4	$0^0 = 1$ encodes this mathematically: potential applied to itself yields the unit

Theorem	A state of pure nothing cannot be maintained. Therefore something must exist.
QED	Something exists because nothing is impossible. [QED]

$1^{(\inf-1)/2n} = 1$ **The potential must manifest. This is not a choice — it is the law of being.**

- Synthesis: Science Answers HOW, T ■ Answers WHY**

Approach	Conclusion	Contribution
Physics (Quantum)	Vacuum fluctuates. Absolute zero energy is impossible.	HOW: the mechanism of non-emptiness
Information Theory	"Nothing" cannot be described without contradiction.	WHAT: the logical impossibility
T0 Theory	NULL is potential, not emptiness. $0^0 = 1$ is inevitable.	WHY: the necessity of manifestation

Synthesis	"Nothing" is impossible: logically, physically, informationally.	THERE IS ONLY SOMETHING.
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- Conclusion**

The question "why does something exist rather than nothing?" has no answer in traditional philosophy because it treats nothing as a valid state. Three independent lines of argument show it is not:

- Physics: quantum laws forbid absolute emptiness
- Information theory: "nothing" requires a medium to be described — contradiction
- T0 Theory: NULL is maximum potential; $0^0 = 1$ is mathematically inevitable

Science describes HOW existence works. T₀ Theory explains WHY it must. Together they give the complete answer: **"nothing" is impossible — logically, physically, and informationally.**

"Something exists because nothing is already a seed.

And a seed cannot fail to sprout."

$1^{(\inf-1)/2n} = 1$ **Axiom.**

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